WEEK 2 LAB TASKS - EXPLORATORY DATA ANALYSIS

CLO - 2: *Design basic AI-based solutions for real-world problems using appropriate techniques such as reasoning, tracking, or robot localization.*

Submission Guidelines:

- Download the .ipynb file from Jupyter Notebook, named it as RollNo_EDA_Name
- Submit **one Pythob Notebook** containing all 12 tasks.

The notebook must include:

  - Clear phase and task headings
  - Python implementation
  - All Task's outputs
  - Short analytical interpretations (where required)

- The notebook should demonstrate **computational reasoning and analytical decision-making**, not only code execution.
- Upload only the single .ipynb file on GCR.

  Important Note:

  Failure to follow the submission instructions **exactly** will result in a **50% deduction marks**.

Problem Context

Artificial Intelligence and Machine Learning systems depend heavily on the quality and structure of the data used to train them.

In real-world applications, data is rarely ready for direct modeling. A dataset may contain missing values, duplicate records, inconsistent categories, skewed distributions, extreme observations, redundant features, or imbalanced target classes.

Exploratory Data Analysis (EDA) provides a systematic way to investigate these issues before model development.

In this lab, you will work with **three real-world datasets** representing different AI applications:

1. **Urban Bike-Sharing Demand Analysis**
2. **Residential House-Price Prediction**
3. **Bank Marketing Response Prediction**

The datasets and scenarios are intentionally different from the examples demonstrated during the lecture. You must therefore apply the EDA concepts learned in the lecture to new real-world situations.

DATASETS USED IN THIS LAB

Dataset 1: Bike Sharing

Real-World Application

A bike-sharing service wants to understand patterns in bicycle rentals and determine which environmental and calendar conditions are associated with higher or lower rental demand.

Official Dataset

Dataset Name: Bike Sharing
Source: UCI Machine Learning Repository
Dataset ID: 275
Link: https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset

The dataset contains hourly and daily bike-rental counts from the Capital Bikeshare system for 2011 and 2012, together with weather and seasonal information. The official UCI repository lists 17,389 instances and 13 features and reports that the dataset has no missing values.

AI Objective

Explore the factors associated with bike-rental demand and determine whether the dataset contains useful patterns for a future demand-prediction system.

Dataset 2:House Prices

Real-World Application

A property analytics company wants to develop an AI system that estimates the selling price of residential properties from their physical and location-related characteristics.

Official Dataset

Dataset Name: House Prices: Advanced Regression Techniques
Source: Kaggle Competition
File: train.csv
Link: https://www.kaggle.com/c/house-prices-advanced-regression-techniques/data

Alternative Link: https://github.com/ankita1112/House-Prices-Advanced-Regression/blob/master/train.csv

The official Kaggle competition provides train.csv, test.csv, and a data-description file. SalePrice is the target variable in the training data.

AI Objective

Investigate the quality, distributions, outliers, and relationships within the property data before developing a house-price prediction model.

Dataset 3:Bank Marketing

Real-World Application

A bank conducts telephone marketing campaigns to contact existing clients and promote term-deposit products.

The bank wants to understand customer characteristics and campaign information associated with whether a client subscribes to a term deposit.

Official Dataset

Dataset Name: Bank Marketing
Source: UCI Machine Learning Repository
Dataset ID: 222
Link: https://archive.ics.uci.edu/dataset/222/bank%2B

The UCI repository describes the dataset as data from direct marketing campaigns conducted through phone calls by a Portuguese banking institution. The classification target is y, which indicates whether the client subscribed to a term deposit (yes/no).

The repository provides several versions. For this lab, use:

bank-full.csv

This version contains the full dataset with 45,211 instances and 17 variables including the target.

AI Objective

Investigate customer and campaign patterns and determine whether the dataset is suitable for developing a classification system that predicts term-deposit subscription.

OVERALL LAB OBJECTIVE

You will perform a complete EDA workflow:

Raw Data

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Phase 1: Understand the Data

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Phase 2: Assess Data Quality

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Phase 3: Understand Distributions

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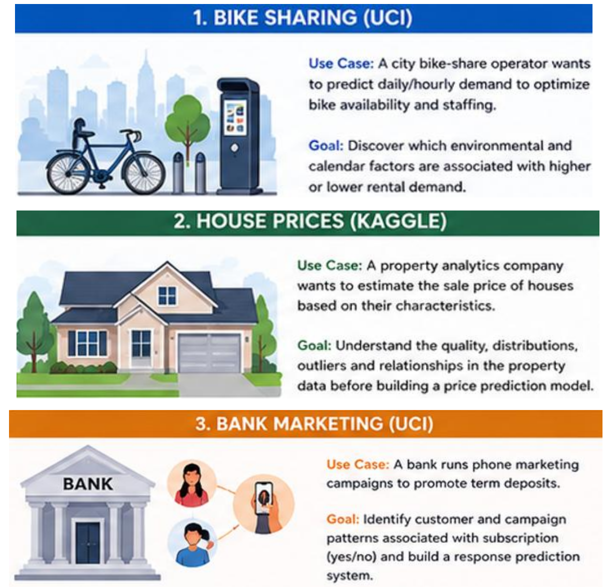
Phase 4: Investigate Outliers

↓

Phase 5: Discover Relationships

↓

Phase 6: Assess AI Readiness



The objective is not merely to execute Python commands. You must explain **what the result means in the corresponding real-world AI scenario**.

You must perform all tasks in a single Jupyter Notebook with proper headings and outputs.

PHASE 1: UNDERSTAND THE DATA

TASK 1: Real-World Dataset Acquisition and Structural Verification

Objective

To perform an initial computational audit of heterogeneous real-world datasets before applying any preprocessing or modeling.

Scenario

You have joined a data science team working with three organizations:

- An urban bike-sharing service
- A property analytics company
- A banking institution

Before building any AI system, your first responsibility is to understand the structure and reliability of the available data.

Instructions

- Load the three datasets into Pandas DataFrames.

For each dataset, perform:

1. Examining sample records from the beginning and end of the dataset

2. Determining dataset dimensionality

3. Inspect attributes

4. Inspect data types and memory usage

5. Inspect missing values

6. Identify potential target and identifier attributes

For example:

- Bike Sharing → cnt
- House Prices → SalePrice
- Bank Marketing → y

Determine whether any attributes represent identifiers rather than meaningful predictive features.

Required Output

Create a summary table:

Dataset	Rows	Columns	Target	Missing Values	Numerical Features	Categorical Features
Bike Sharing						
House Prices						
Bank Marketing						

Analytical Questions

1. What type of data does each organization collect?
2. Which dataset contains the greatest variety of attribute types?
3. Which columns appear to be identifiers?
4. Which dataset is intended for regression?
5. Which dataset is intended for classification?
6. Does the absence of missing values automatically mean that a dataset is ready for AI?

TASK 2: Attribute Role, Data Type and Feature Cardinality Analysis

Objective

To classify attributes according to their computational role and identify features that may require special preprocessing.

Scenario

Before feeding data into an AI model, a data scientist must understand what each column represents.

Using the **Bank Marketing** dataset, construct a DataFrame that classifies each attribute as:

- Numerical Feature
- Categorical Feature
- Binary Feature
- Target Variable
- Identifier / Administrative Attribute

Example:

```
attribute_roles = pd.DataFrame({
    "Attribute": [...],
    "Data_Type": [...],
    "Role": [...]
})
attribute_roles
```

Feature Cardinality

- Calculate the number of unique values: e.g. bank_df.nunique()
- Investigate attributes such as: bank_df["job"].nunique()
- **Investigate Categorical Values e.g.** bank_df["job"].value_counts()

Important Question

Suppose an attribute contains almost one unique value for every customer.

Ask:

Would this attribute provide useful predictive information, or is it simply identifying individual records?

Explain your reasoning.

Additional Investigation

The Bank Marketing dataset contains categorical values such as "unknown" for some attributes. Do **not automatically treat "unknown" as a numerical missing value**.

Investigate how frequently "unknown" occurs and decide whether it should be treated as:

- A meaningful category
- A missing-information indicator
- A value requiring further investigation

Justify your decision.

PHASE 2: ASSESS DATA QUALITY

TASK 3 : Missing Data Investigation and Imputation

Objective

To identify missing values in a real-world dataset and apply simple imputation techniques to prepare the data for AI/ML analysis.

Scenario

A wildlife research team has collected measurements of penguins from different islands. Some measurements are missing because they were not successfully recorded during field observations.

Before developing a penguin-species classification system, the data scientist must identify and handle the incomplete records.

Dataset

Use the **Palmer Penguins dataset** available directly through Seaborn.

Step 1: Load the dataset

Step 2: Inspect the dataset

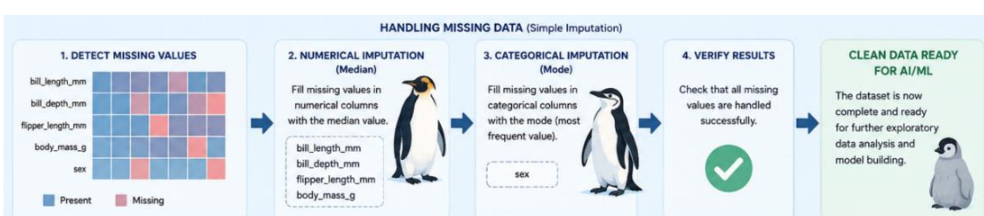
Step 3: Find missing values

Step 4: Handle numerical missing values

- **For numerical attributes, use the median**

Step 5: Handle categorical missing values

- **For sex, use the mode**



Step 6: Verify

Attribute	Missing Before	Missing %	Type	Imputation
bill_length_mm	...	...	Numerical	Median
bill_depth_mm	...	...	Numerical	Median
flipper_length_mm	...	...	Numerical	Median
body_mass_g	...	...	Numerical	Median
sex	...	...	Categorical	Mode

Important Analytical Question

1. Which attributes contained missing values?
2. Which attribute had the highest number of missing values?
3. Why did we use the **median** for numerical attributes?
4. Why did we use the **mode** for sex?
5. How many missing values remained after imputation?
6. Why is handling missing data important before training an AI model?

TASK 4: Duplicate, Consistency and Data-Integrity Check

Objective

To identify redundant observations and potential inconsistencies that could distort analysis or model training.

Part A: Duplicate Detection

- Perform duplicate detection for all three datasets:
- If duplicate rows are found, drop them
- Document the before and after dataset shape.

If no duplicates are found, report that explicitly rather than artificially creating duplicates.

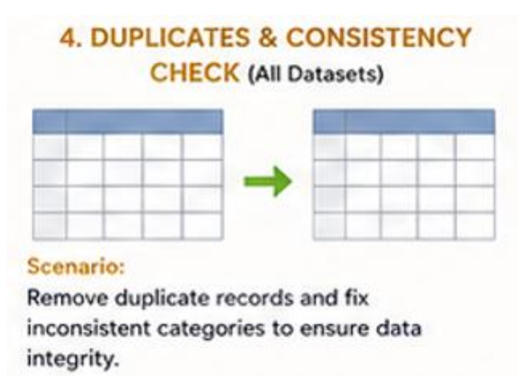
Part B: Categorical Consistency

Using the **Bank Marketing** dataset, inspect:

bank_df["job"].value_counts()

bank_df["marital"].value_counts()

bank_df["education"].value_counts()



bank_df["poutcome"].value_counts()

Look for:

- Unexpected labels
- Spelling differences
- Leading/trailing spaces
- Duplicate representations of the same category

If necessary, remove leading and trailing spaces and rectify other inconsistencies.

Part C: Data-Type Verification

- Verify that numerical variables are actually numerical.e.g. .info()

Analytical Question

How can duplicate records or inconsistent categories affect an AI model even when the Python code executes without an error?

PHASE 3: UNDERSTAND DISTRIBUTIONS

TASK 5: Descriptive Statistical and Group-Level Analysis

Objective

To characterize numerical variables and compare real-world groups using descriptive statistics.

Part A: Bike-Sharing Demand

A city transportation team wants to understand whether bike demand differs between working and non-working days.

Computationally summarize numerical features and analyze group-level statistical behavior and calculate median demand as well.

Question

Is bike demand substantially different between working and non-working days?

Part B: House Prices

Computationally summarize numerical features and analyze group-level statistical behavior.

For SalePrice calculate: mean, median, standard deviation, minimum, and maximum.

Also calculate statistics for: GrLivArea, GarageArea, OverallQual

Question

If the mean sale price is substantially higher than the median, what might this suggest about the distribution?

TASK 6: Distributional Visualization and Pattern Discovery

Objective

To visualize distributions and identify patterns that may not be obvious from numerical summaries.

Generate Histograms

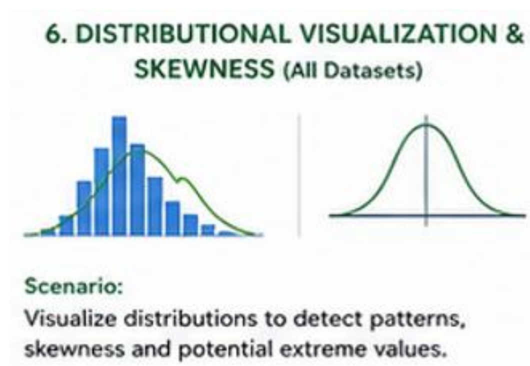
Bike Sharing

- Create histograms with 30 bins and Kernel density for:
    - temp
    - hum
    - cnt
- Label x-axis as "Total Bike Rentals" and y-axis as “Frequency”
- Set the title of the graph as "Distribution of Bike Rental Demand"

House Prices

Create histograms for:

- SalePrice



Analyze Each Distribution

For every selected variable, comment on:

- Center
- Spread
- Symmetry
- Skewness
- Concentration
- Potential extreme observations

Skewness

- Calculate: “SalePrice” & “temp” skewness
- Interpret the result.

Analytical Question

Why is visualization sometimes more informative than simply reporting the mean?

PHASE 4: INVESTIGATE OUTLIERS

TASK 7: Statistical Outlier Identification Using IQR

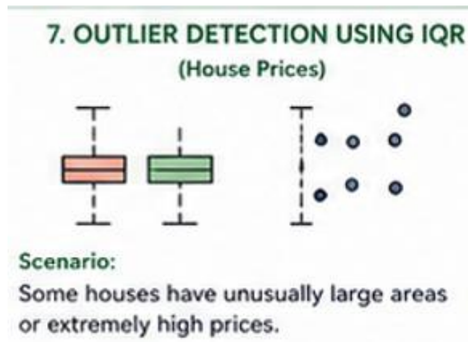
Objective

To identify potentially anomalous observations using a robust statistical method.

Scenario

The property analytics company discovers that some houses have unusually large living areas and exceptionally high prices.

Before building a price-prediction model, investigate whether these observations are errors or legitimate extreme properties.



Use:

- SalePrice
- GrLivArea
- GarageArea

Calculate

- Q1
- Q3
- IQR
- Lower bound
- Upper bound
- Number of outliers

Generate Box Plots

Create box plots for:

- SalePrice
- GrLivArea
- GarageArea

Important Real-World Question

Does an IQR outlier automatically mean that the observation is wrong?

Discuss whether an unusually expensive house could be a legitimate observation.

Requirement

Do not remove any observations in this task.

TASK 8: Outlier Mitigation Through IQR-Based Filtering

Objective

- To investigate the effect of removing statistically identified extreme observations.
- Assume that the organization has determined that the detected extreme observations represent erroneous records and should be removed for this particular modeling pipeline.
- Using the bounds from Task 7, create a cleaned dataset. i.e. clean_house_df (For this task, use the IQR bounds of SalePrice and GrLivArea to create the cleaned dataset., remove observations that fall outside the corresponding bounds and create clean_house_df).
- Document: Original shape i.e. **house_df.shape** & Cleaned shape i.e. **clean_house_df.shape**
- Calculate the percentage of observations removed.

Analytical Question

What information could be lost if legitimate extreme properties wer
were statistical outliers?

TASK 9: Post-Processing Statistical Reassessment

Objective

To determine how outlier mitigation changes the statistical characteristics of the dataset.

Compare **before vs after** outlier removal. Also Calculate the before and after SalePrice Range.

Calculate:

- Mean
- Median
- Standard deviation
- Minimum
- Maximum

Example:

```
comparison = pd.DataFrame({
    "Before": [
        house_df["SalePrice"].mean(),
…
    ],
    "After": [
        clean_house_df["SalePrice"].mean(),
        …
    ]})
comparison.index = [
    "Mean",
    …]
comparison
```

Analytical Questions

1. Did the mean change?
2. Did the median change?
3. Did the standard deviation decrease?
4. Did the range become narrower?
5. Did the distribution become more stable?
6. Does removing outliers necessarily make the dataset better?

PHASE 5: DISCOVER RELATIONSHIPS

TASK 10: Bivariate Relationship Analysis

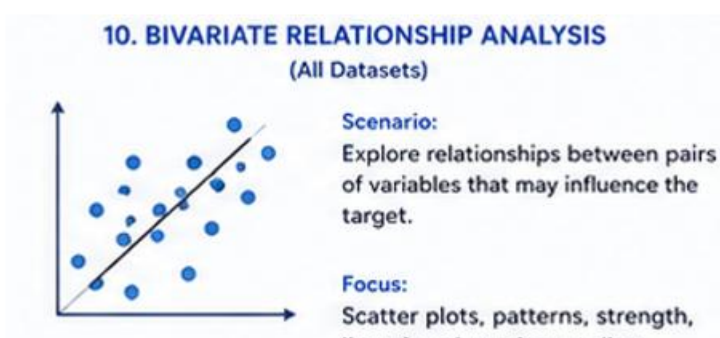
Objective

To identify relationships between variables that may be useful for future AI models.

House Prices Dataset

Generate scatter plots for:

- GrLivArea vs SalePrice



- GarageArea vs SalePrice
- YearBuilt vs SalePrice

Also calculate the numerical correlation and

Visualize through the heatmap.

Analytical Question

Which property characteristic appears most strongly associated with sale price based on the visualization?

Use scatter plots and discuss whether the variables show:

- Strong relationship
- Moderate relationship
- Weak relationship
- No obvious relationship
- Clusters

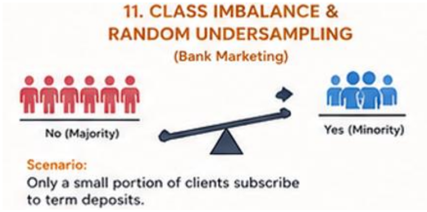
TASK 11: Class Imbalance Detection and Corrective Sampling

Objective

To identify class imbalance in a real-world AI classification problem and investigate the effect of corrective sampling.

Scenario

The bank wants to develop an AI system that predicts whether a client will subscribe to a term deposit.



The target variable is:

y = yes

y = no

Because successful subscriptions may be less frequent than non-subscriptions, investigate whether the target distribution is balanced.

Step 1: Calculate Class Frequencies e.g. bank_df["y"].value_counts()

Step 2: Calculate Percentages

Step 3: Visualize via countplot

- Label x-axis ("Term Deposit Subscription")
- Label y-axis ("Number of Clients")
- Set title("Bank Marketing Target Distribution")

Step 4: Apply Random Undersampling (Think! should we Apply this technique here?) if not just skip this procedure(i.e. step 4-6) and write the reason/justification.

Step 5: Verify the class balance by before and after bank_df shape.

Step 6: Generate a post-balancing count plot.

Analytical Questions

1. Was the target balanced?
2. Which class was the majority?
3. How many observations were removed?
4. What information may have been lost through undersampling?
5. Why can a highly imbalanced dataset make accuracy misleading?
6. Is random undersampling always the best solution?

TASK 12: Correlation Analysis and Feature Redundancy

Objective

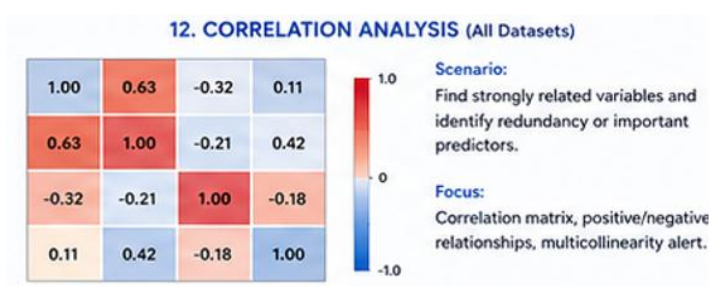
To identify statistically related numerical features and investigate potential redundancy.

Part A: Bike Sharing

- Calculate correlation of numerical features
- Visualize heatmap with annot=True and cmap=”coolwarm”, set title as "Bike Sharing Correlation Matrix".

Identify:

- Strong positive correlation
- Strong negative correlation, if present
- Features strongly related to cnt
- Highly correlated feature pairs



Part B: House Prices

- Calculate correlation
- Visualize the correlation matrix.

- Identify the numerical variables most strongly correlated with: SalePrice
- Because House Prices has many numerical features so display first 10 correlated features with salePrice then last 5.

sImportant Warning

Correlation measures statistical association between numerical variables. A strong correlation does **not** prove that one variable causes another.

Analytical Questions

1. Which relationships are strongest?
2. Which features appear potentially redundant?
3. Which features show strong association with the target or relevant outcome?
4. Can correlation alone determine which features should be removed?
5. Why should domain knowledge also be considered?

UNGRADED ACTIVITY:

PHASE 6: THINK LIKE A DATA SCIENTIST

TASK 13: Structured EDA and AI-Readiness Report

Objective

To consolidate the complete EDA process into an analytical assessment of whether each dataset is suitable for subsequent AI/ML modeling.

You are now reporting your findings to the AI development team.

The team does not want a list of Python commands.

They want to know:

What did we discover about the data, what problems did we solve, and what must happen before model development?

Part A: Dataset Comparison

Create:

Dataset	Major Data Issues	Cleaning Performed	Important Patterns	Modeling Concerns	AI Readiness
Bike Sharing					

House Prices					
Bank Marketing					

Part B: Feature Assessment

For each dataset, identify:

Useful Features

Identify features that appear potentially useful based on your analysis.

Potentially Irrelevant Features

Identify attributes that may not provide meaningful predictive information.

Examples may include identifiers or administrative fields.

Features Requiring Further Preprocessing

Identify features requiring:

- Imputation
- Encoding (binary, label, one-hot)
- Transformation
- Scaling
- Outlier treatment
- Feature selection
- Other preprocessing

Part C: Target Analysis

For each dataset, identify:

Bike Sharing

Prediction target: cnt

Task type: Regression

House Prices

Prediction target: SalePrice

Task type: Regression

Bank Marketing

Prediction target: y

Task type: Binary Classification

Explain why each target requires a different type of AI/ML modeling approach.

Part D: Data Leakage Awareness

For each dataset, inspect whether any feature could potentially contain information that would not be available at the time the prediction is actually made.

Ask:

- Would this information genuinely be available when the AI system makes its prediction?
- If you identify a potentially problematic feature, explain why.
- If you do not identify obvious leakage, state that and explain your reasoning.

Example: Take Bank Marketing duration and prediction timing as a data-leakage example.

Final Decision Question

Among the three datasets, which one requires the most careful preprocessing before AI/ML model development, and what specific evidence from your EDA supports your decision?

Your answer must be based on your computed results and visualizations rather than general assumptions.